Blockchain-Enabled Intrusion Detection and Prevention System of APTs Within Zero Trust Architecture

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Mohamed Shahal

Univ. Reg.No. VAS23CS133

(AN ISO 9001:2008 CERTIFIED INSTITUTION)

(Accredited by NAAC with "B++" grade)

Department of Computer Science and Engineering

Vidya Academy of Science & Technology

Thalakkottukara, Thrissur - 680 501

(http://www.vidyaacademy.ac.in)

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Department of Computer Science and Engineering
Vidya Academy of Science & Technology

Thalakkottukara, Thrissur - 680 501

(http://www.vidyaacademy.ac.in)

(AN ISO 9001:2008 CERTIFIED INSTITUTION)

Certificate

This is to certify that the seminar report titled **Blockchain-Enabled Intrusion Detection and Prevention System of APTs Within Zero Trust Architecture** is a bonafide record of the work carried out by **Mohamed Shahal (Univ.Reg.No. VAS23CS133)** of Vidya Academy of Science & Technology, Thalakkottukara, Thrissur - 680 501 in partial fulfillment of the requirements for the award of **Degree of Bachelor of Technology** in **Computer Science and Engineering** of **APJ Abdul Kalam Technological University**, during the academic year 2023-2027.

Seminar Guide

Dr. Sajay K R

Professor and HOD, Dept. of CSE

Head of Department

Dr. Sajay K R

Professor, Dept. of CSE

Seminar Coordinators

Dr. Sajay K R

Professor, Dept. of CSE

Ms. Reshma V M

Assistant Professor, Dept. of CSE

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Mohamed Shahal
Reg. No. VAS23CS133
Seventh Semester B.Tech
Vidya Academy of Science & Technology
Thrissur - 680 501.

October 2026

Abstract

Modern organizations rely on remote work, cloud virtualization and Bring Your Own Device (BYOD) policies, which dissolve the traditional network perimeter. Zero Trust Architecture (ZTA) responds by trusting no identity, endpoint or session by default, yet it has an Achilles' heel: once a remote user establishes an authenticated session, the endpoint itself becomes a trusted tunnel into the corporate network. If an Advanced Persistent Threat (APT) has silently taken root on that device, its malware can piggyback on the trusted channel. Existing intrusion detection systems struggle against Living-off-the-Land techniques that abuse legitimate tools such as PowerShell, and blockchain-based collaborative proposals are largely theoretical. This report studies a Blockchain-Enabled Intrusion Detection and Prevention System (BIDPS) that protects endpoints within a ZTA. The system registers the SHA-512 hash of every legitimate executable on a permissioned Hyperledger Fabric ledger through Node.js smart contracts, and maps each resource to an owner identity so that high-risk binaries are locked to administrator contexts. Before any program runs, its hash and owner are verified against the ledger, and any non-whitelisted or un-owned binary is blocked and an alert is raised. The framework was evaluated on a virtual testbed against 46 MITRE ATT&CK techniques emulated with CALDERA, achieving a 100% success rate for detecting and preventing file-based attacks. For fileless attacks, success was 63.04% without memory analysis and rose to 84.7% when Sysmon Event ID 8 logs were integrated, giving an overall defense success of 95.2%. The work shows that blockchain can serve as an active validation gate and immutable source of truth rather than a passive log.

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Vision of the Institution

To be a premier institution in engineering education and research, nurturing innovative professionals for sustainable societal development.

Mission of the Institution

M1. Provide quality engineering education that empowers students with strong technical competence and analytical problem solving abilities.

M2. Promote research, innovation, entrepreneurship and industry collaboration to develop sustainable technological solutions.

M3. Develop ethical and socially responsible professionals with leadership qualities, lifelong learning abilities and a commitment to societal well-being.

Vision of the Department

To excel in computer science and engineering education and research, nurturing innovative and ethically responsible professionals for sustainable technological development.

Mission of the Department

M1. Impart quality education in computer science and engineering by providing strong theoretical foundations, practical skills and exposure to emerging technologies.

M2. Promote research, innovation, entrepreneurship and industry collaboration to develop sustainable and socially relevant computing solutions.

M3. Develop graduates with professional competence, ethical values, logical and analytical thinking skills, leadership qualities and a commitment to lifelong learning and societal well-being.

Program Educational Objectives (PEOs)

PEO-1: Graduates shall practice computer science and engineering in IT industry, public sector undertaking, as an entrepreneur or as a researcher pursuing a successful professional career.

PEO-2: Graduates shall demonstrate creativity, develop innovative ideas and work in teams to accomplish a common goal.

PEO-3: Graduates shall demonstrate project management skills, lifelong learning and social awareness upholding ethical values.

Program Outcomes (POs)

PO1	**Engineering Knowledge:** Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.
PO2	**Problem Analysis:** Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)
PO3	**Design/Development of Solutions:** Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for public health, safety, and environment. (WK5)
PO4	**Conduct Investigations of Complex Problems:** Conduct investigations using research-based knowledge including design of experiments, modelling, analysis and interpretation of data. (WK8)
PO5	**Engineering Tool Usage:** Create, select and apply appropriate techniques and modern engineering tools recognizing their limitations. (WK2 and WK6)
PO6	**The Engineer and The World:** Analyze societal and environmental impacts with reference to sustainability, economy, health, safety, and culture. (WK1, WK5, WK7)
PO7	**Ethics:** Apply ethical principles, professional ethics, and adhere to laws. (WK9)
PO8	**Individual and Collaborative Team Work:** Function effectively as an individual and in teams.

PO9	**Communication:** Communicate effectively through reports, documentation, and presentations considering diverse audiences.
PO10	**Project Management and Finance:** Apply management principles and economic decision-making in projects.
PO11	**Life-Long Learning:** Recognize the need for independent and lifelong learning. (WK8)

Knowledge and Attitude Profile (WK)

WK1	A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.
WK2	Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.
WK3	A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.
WK4	Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.
WK5	Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.
WK6	Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.

WK7	Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.
WK8	Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.
WK9	Ethics, inclusive behavior and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.

Program Specific Outcomes (PSOs)

PSO-1	The ability to apply the knowledge of computational structures and algorithms for solving complex engineering problems in multidisciplinary areas.
PSO-2	The ability to design, develop and evaluate computer-based systems using advanced tools and technologies to meet real-world challenges.

Sustainable Development Goals (SDGs)

SDG 1	**No Poverty:** End poverty in all its forms everywhere.
SDG 2	**Zero Hunger:** End hunger, achieve food security and improved nutrition and promote sustainable agriculture.
SDG 3	**Good Health and Well-being:** Ensure healthy lives and promote well-being for all at all ages.
SDG 4	**Quality Education:** Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.
SDG 5	**Gender Equality:** Achieve gender equality and empower all women and girls.
SDG 6	**Clean Water and Sanitation:** Ensure availability and sustainable management of water and sanitation for all.
SDG 7	**Affordable and Clean Energy:** Ensure access to affordable, reliable, sustainable and modern energy for all.
SDG 8	**Decent Work and Economic Growth:** Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all.
SDG 9	**Industry, Innovation and Infrastructure:** Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation.
SDG 10	**Reduced Inequalities:** Reduce inequality within and among countries.
SDG 11	**Sustainable Cities and Communities:** Make cities and human settlements inclusive, safe, resilient and sustainable.
SDG 12	**Responsible Consumption and Production:** Ensure sustainable consumption and production patterns.

SDG 13	**Climate Action:** Take urgent action to combat climate change and its impacts.
SDG 14	**Life Below Water:** Conserve and sustainably use the oceans, seas and marine resources for sustainable development.
SDG 15	**Life on Land:** Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss.
SDG 16	**Peace, Justice and Strong Institutions:** Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels.
SDG 17	**Partnerships for the Goals:** Strengthen the means of implementation and revitalize the global partnership for sustainable development.

SDGs addressed with Justification

Seminar Title	**SDG addressed**	**Justification**
Blockchain-Enabled IDPS for APTs within Zero Trust Architecture	SDG 9: Industry, Innovation and Infrastructure	Presents a blockchain-based defense that strengthens the cyber resilience of enterprise and critical digital infrastructure against Advanced Persistent Threats. This is the primary goal addressed.
	SDG 4: Quality Education	Protects endpoints used for remote and hybrid learning so academic systems and student records stay secure.
	SDG 3: Good Health and Well-being	Protects hospital and health service endpoints that hold sensitive patient data from targeted attacks.

POs & PSOs addressed with Justification

Seminar Title	**POs & PSOs addressed**	**Justification**
Blockchain-Enabled IDPS for APTs within Zero Trust Architecture	PO1 – Engineering Knowledge	Applies hashing (SHA-512), distributed ledgers, network security and Zero Trust principles to endpoint protection.
	PO2 – Problem Analysis	Analyzes the weakness of trusted endpoint sessions, Living-off-the-Land attacks and the lack of immutable trust in distributed IDS.
	PO3 – Design/Development of Solutions	Studies a design of whitelisting, smart contracts, user-context isolation and a verification guard that blocks untrusted execution.
	PO4 – Conduct Investigations of Complex Problems	Reviews the reference IEEE Access study and interprets its results across 46 MITRE ATT&CK techniques.
	PO5 – Modern Tool Usage	Examines Hyperledger Fabric, Node.js contracts, CALDERA, Sysmon and iptables, and notes limits for fileless attacks.
	PO6 – The Engineer and The World	Reduces the societal and economic impact of cyber attacks on organizations that depend on remote work.

Seminar Title	**POs & PSOs addressed**	**Justification**
	PO7 – Ethics	Verifies software against an immutable ledger, records block events transparently and avoids overstated claims.
	PO8 – Individual and Team Work	Works independently and incorporates feedback from the guide, coordinators and peers during the seminar.
	PO9 – Communication	Communicates the study through this report and a seminar presentation.
	PO10 – Project Management and Finance	Plans the seminar within the schedule and considers the infrastructure cost of ledger and monitoring.
	PO11 – Life-Long Learning	Studies emerging security topics independently, including Zero Trust, blockchain and APT behaviour.
	PSO1	Applies hashing, ledger lookup and access rules to decide whether a program may execute.
	PSO2	Evaluates a testbed-based system using advanced tools to meet real-world security challenges.

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List of Abbreviations

ZTA	Zero Trust Architecture
BIDPS	Blockchain-Enabled Intrusion Detection and Prevention System
APT	Advanced Persistent Threat
SDP	Software-Defined Perimeter
SPA	Single Packet Authorisation
BYOD	Bring Your Own Device
LotL	Living-off-the-Land
IDS	Intrusion Detection System
IDPS	Intrusion Detection and Prevention System
SHA	Secure Hash Algorithm
C2	Command and Control
MITRE ATT&CK	Adversarial Tactics, Techniques and Common Knowledge

Chapter 1

INTRODUCTION

1.1 Background and Motivation

Organizational IT infrastructures are changing rapidly. Remote working, cloud virtualization and Bring Your Own Device (BYOD) have become the new normal, and employees now reach corporate resources from home networks and personal laptops. Traditional security was built around a static perimeter defended by firewalls, web application firewalls (WAFs) and virtual private networks (VPNs). This model fails in the modern setting because endpoints routinely bypass the boundary altogether.

Zero Trust Architecture (ZTA) replaces perimeter trust with a default breach mindset: no identity, endpoint, transaction or session is implicitly trusted, and continuous cryptographic validation is required at every dynamic access point. A widely used way of realizing ZTA is the **Software-Defined Perimeter (SDP)**, in which endpoints are treated as individual islands of isolated resources. A remote client first sends an encrypted and signed UDP Single Packet Authorisation (SPA) packet. The inbound firewall drops everything by default, so the network is blind to unauthorized scans, and ports are opened temporarily for the verified client only (Rose et al., 2020).

1.2 Technological Advancements in Endpoint Defense

Endpoint defense has progressed from signature-based antivirus to intrusion detection and prevention systems (IDPS) that monitor system calls, logs and network traffic. Collaborative and distributed IDSs share alerts among several hosts to improve detection,

but they face coordination and trust problems: a compromised participant can inject false data or suppress genuine alerts.

Blockchain offers a way to remove this trust problem. A permissioned ledger such as **Hyperledger Fabric** records data immutably and replicates it across peers using consensus, while smart contracts (chaincode), here written in Node.js, execute business logic on that ledger. Because policy is stored on-chain, an attacker who compromises a single node cannot rewrite it.

Recent proposals combine blockchain with intrusion detection, but most use the ledger only as a passive store for alerts and logs, and many are theoretical with no implementation details. The system studied in this report instead turns the blockchain into an active decision point that every program execution must pass.

1.3 Role of Blockchain in Zero Trust Endpoint Protection

The weak spot of a ZTA is the authenticated endpoint. APTs are stealthy, long-term attacks that infiltrate a device through phishing documents, macros or vulnerable software and then live off the land, using trusted tools such as PowerShell and `cmd.exe` to avoid signature detection. Once such malware sits on a device with an open, trusted session, the whole network is exposed.

BIDPS addresses this with three ideas. *Hash-based whitelisting* registers every legitimate executable using SHA-512. *Explicit on-chain verification* moves the trust decision out of the endpoint's own control and into a distributed, tamper-proof ledger. *User-context isolation* maps asset ownership to user sessions so that high-risk binaries, for example administrative tools, run only in an administrator context, which neutralizes fileless execution from standard user channels.

1.4 Goals and Purposes

The overarching goal of this study is to protect endpoints inside a Zero Trust Architecture from Advanced Persistent Threats by turning blockchain from a passive logging tool into an active validation and execution gate.

The first objective is to identify and register every legitimate executable and binary extension using SHA-512 hashes on a permissioned blockchain. The second is to verify execution rights against the ledger in real time, before a program is launched, and to block and alert on any non-whitelisted or un-owned binary.

The third objective is to break the attack chain early, before the lateral movement stage, so that security is proactive rather than reactive. The fourth is to evaluate the framework experimentally against realistic APT behaviour emulated with the CALDERA framework and mapped to MITRE ATT&CK techniques, and to measure both file-based and fileless performance.

Chapter 2

LITERATURE REVIEW

2.1 LITERATURE SURVEY

2.1.1 Zero Trust Architecture and Software-Defined Perimeter

- **Authors:** Rose et al., 2020
- **Source:** NIST Special Publication 800-207

Rose et al. (2020) define Zero Trust Architecture as a set of principles in which trust is never granted based on network location alone and every access request is authenticated, authorized and continuously evaluated. Enclave-based deployments such as the Software-Defined Perimeter hide protected resources behind a drop-all firewall and open them only after a signed SPA packet from a verified client.

The approach greatly reduces the externally visible attack surface. However, it verifies the user and device at connection time and does not inspect what runs on the endpoint afterwards. An APT already present on the device can therefore ride on the trusted channel, which is the gap addressed by BIDPS.

2.2 LITERATURE SURVEY

2.2.1 Collaborative Intrusion Detection Systems

- **Focus:** Distributed and collaborative IDS for coordinated detection

- **Limitation:** Trust among cooperating nodes

Collaborative IDS designs let several hosts or sensors exchange alerts and signatures so that an attack seen at one site can be detected at others. The studies show improved detection coverage over isolated sensors.

Their main weakness is coordination trust. Participants must rely on the integrity of shared data, and a compromised or malicious member can poison it. Signature-based detection also remains weak against Living-off-the-Land techniques that use legitimate system tools to evade detection.

2.3 LITERATURE SURVEY

2.3.1 Blockchain-Enabled Intrusion Detection and Prevention of APTs within ZTA

- **Source:** IEEE Access, 2022
- **Focus:** Hash-based on-chain whitelisting with Hyperledger Fabric

The reference paper for this seminar, published in IEEE Access (2022), proposes BIDPS for endpoint protection within a ZTA. It registers SHA-512 hashes of legitimate binaries and their owners on a Hyperledger Fabric network and checks every execution request against the ledger. The evaluation uses a virtual testbed with the CALDERA emulation framework and 46 MITRE ATT&CK techniques drawn from APT behaviour.

The work shows that a blockchain-based system can be built and measured, not just proposed, and reports full success against file-based attacks. Its stated limitation is fileless, memory-resident attacks, which do not create new files and therefore need additional telemetry such as Sysmon events. Future work targets writing kernel-level event identifiers to the chain and large-scale throughput testing.

Chapter 3

COMPONENTS OF THIS SYSTEM

3.1 Zero Trust Enclave Gateway (SDP)

The first component is the Zero Trust gate, an Ubuntu firewall running iptables in default-drop mode. A remote endpoint client sends a signed SPA packet over UDP; the controller gateway verifies the policy; and only then does the firewall change state and open ports temporarily for the verified target address. This keeps the network invisible to unauthorized scans.

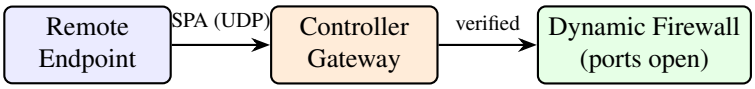


Figure 3.1: Enclave-based software-defined perimeter flow

3.2 Hash-Based Whitelisting Module

Every legitimate executable and binary extension on the endpoint is hashed with SHA-512, a high collision-resistance algorithm, using a file hashing utility. The resulting hash strings form the whitelist. Initial hashing of all valid installed binaries takes about 52.83 seconds. A program whose hash is not on the whitelist is prevented from running immediately.

3.3 Blockchain Ledger and Smart Contract Layer

A permissioned Hyperledger Fabric v2.2.2 network runs on an Ubuntu ledger peer. Node.js smart contracts expose the functions listed in Table 3.1. Each ledger record holds the resource identifier, SHA-512 hash, file size, assigned owner identity and a resource version bound.

Table 3.1: Smart contract functions and ledger record fields

Function	**Record Field**	**Outcome**
InitLedger()	Resource ID	Initializes the trusted base endpoint configuration
CreateResource()	SHA-512 hash string	Appends newly compiled, trusted patch binary signatures
ResourceExists()	File size (bytes)	Real-time check before a local program launches
UpdateResource()	Assigned owner identity	Alters security version bounds on administrative updates
TransferResource()	Resource version bound	Granular, dynamic ownership transfer

3.4 User-Context Isolation and Verification Guard

Asset ownership is mapped to active user sessions. High-risk processes such as administrative binaries are locked to an Admin owner context. The verification guard intercepts each execution request and applies the access rule

$$\text{VerifyAccess}(R_i, U_i) = \begin{cases} \text{ALLOW} & \text{if } \text{SHA512}(R_i) \in H_{ledger} \wedge \text{Owner}(R_i) = U_i \\ \text{BLOCK \& ALERT} & \text{otherwise} \end{cases}$$

where R_i is the requested resource, U_i the requesting user and H_{ledger} the set of hashes on the ledger.

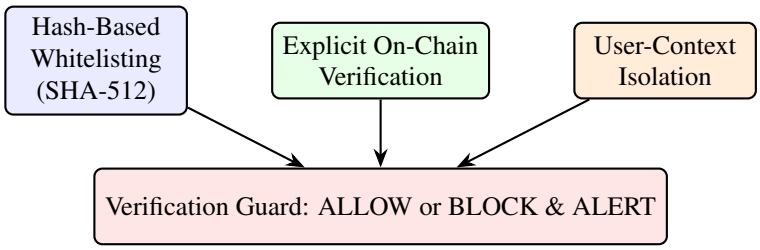


Figure 3.2: BIDPS mitigation framework

Chapter 4

METHODOLOGY

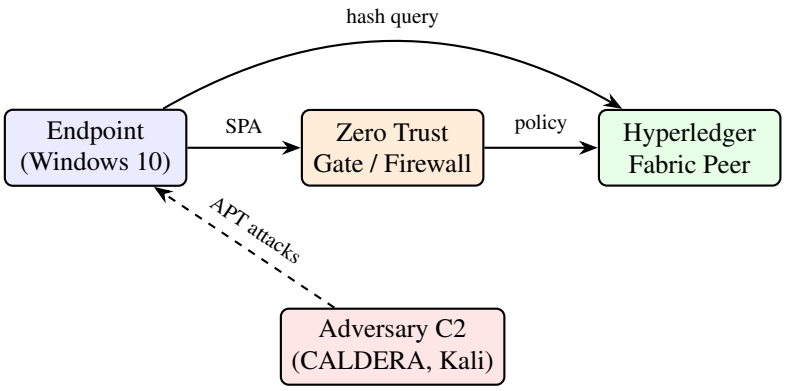


Figure 4.1: Working Flow Chart

4.1 Working Mechanism

4.1.1 Working Mechanism

BIDPS operates as a gate between a request to run a program and the operating system. At enrolment the endpoint is registered; hashes of valid binaries are synchronized to the ledger; and from then on every execution request is intercepted and checked against the ledger. Because the ledger is permissioned and replicated, the policy cannot be modified from a compromised endpoint.

4.1.2 Endpoint Enrolment

The user generates a cryptographic wallet, device keys and X.509 authorization certificates. These identify the user and the device to the Hyperledger Fabric network and tie later ownership records to a verified identity.

4.1.3 Binary Hash Setup and On-Chain Sync

SHA-512 hashes of all valid installed binary assets on disk are generated, which takes roughly 52.83 seconds for the initial setup. The hashes are then synchronized to the permissioned ledger nodes through `InitLedger()` and `CreateResource()` smart contract calls, producing the trusted baseline.

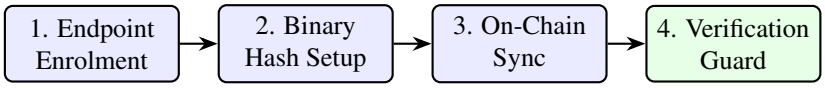


Figure 4.2: BIDPS system execution sequence

4.1.4 Verification Guard

When a program is launched, the guard intercepts the request in real time, computes the SHA-512 of the binary and queries `ResourceExists()`. The request is allowed only if the hash is on the ledger and the owner matches the user context. A non-whitelisted or un-owned binary triggers an immediate BLOCK and an administrator alert.

4.1.5 Handling Fileless Attacks

Fileless injections, such as APT37 ROKRAT memory injection through CVE-2018-0802, bypass filesystem checks because no new file is written. To address them, Sysmon Event ID 8 (CreateRemoteThread) logs are integrated with the BIDPS consensus verification, which reduces successful memory evasion.

4.1.6 Experimental Testbed

The evaluation used four virtual machines, summarized in Table 4.1. The attacks were emulated with the CALDERA framework and cover 46 MITRE ATT&CK techniques across Initial Access, Execution, Persistence, Privilege Escalation, Defense Evasion, Credential Access and Discovery. All attacks were executed before the lateral movement stage to demonstrate proactive security.

4.1.7 File-Based Attack Detection Results

BIDPS blocked every file-based technique tested. In the APT30 StickyNotes vector the dropped payload was blocked because its SHA-512 hash was not on-chain. In the APT29 Office macro case, encoded JSE files were refused because the scripting inter-

Table 4.1: Evaluation testbed specifications

Role	CPU and RAM	OS	Software
Remote Endpoint VM	Intel Core i7-8750H, 8 GB	Windows 10 Enterprise	HashMyFiles, CALDERA agent, Sysmon
Blockchain Ledger Peer	4 vCPUs, 8 GB	Ubuntu 20.04 LTS	Hyperledger Fabric v2.2.2, Node.js contracts
Adversary C2 VM	2 vCPUs, 4 GB	Kali Linux 2021.2	CALDERA server, reverse SSH, Empire
Zero Trust Gate / Firewall	2 vCPUs, 2 GB	Ubuntu 20.04 LTS	iptables default-drop, SPA verification

preter was restricted for standard user contexts. In the APT41 registry DLL case, the malicious DLL mapping and `regsvr64` execution were denied. A privilege escalation attempt using Akagi64 and WOW64logger path hijacking failed because the custom binary had no ledger entry, and the credential access attempt using a SAM dumping tool was denied, with Admin-owned PowerShell preventing scripts such as PowerDump. This gave a 100% success rate for the 46 file-based techniques.

4.1.8 Performance Under APT Attacks

Results are summarized in Table 4.2 and Figure 4.3. Without a memory analyzer, fileless attacks were stopped in 63.04% of cases; adding Sysmon Event ID 8 integration raised this to 84.7%, and the overall defense success of the hybrid framework reached 95.2%.

Table 4.2: BIDPS success rates under APT attacks

Attack Category	Success Rate
File-based vectors (46/46)	100%
Fileless (no memory analyzer)	63.04%
Fileless (with Sysmon integration)	84.7%
Overall hybrid security integrity	95.2%



Figure 4.3: Success rates under APT attacks

Chapter 5

APPLICATION

5.1 Remote and Hybrid Workforce Security:

Employees working from home or on personal devices can be admitted into a ZTA while BIDPS ensures that only whitelisted software runs on their endpoints, even when the session is trusted.

5.2 Enterprise Endpoint Protection:

Organizations can maintain a ledger of approved binaries for all workstations and block unknown executables, reducing the impact of phishing documents, malicious macros and dropped payloads.

5.3 Defense Against Living-off-the-Land Attacks:

Locking high-risk tools such as PowerShell to administrator contexts prevents their misuse from standard user sessions, neutralizing a common APT evasion technique.

5.4 Critical Infrastructure and Government Networks:

Environments that face targeted, long-term attacks benefit from an immutable policy store that a compromised host cannot alter.

5.5 Software Update and Patch Management:

Trusted patches are added to the ledger through `CreateResource()` and `UpdateResource(` so legitimate updates run while unapproved modified binaries are rejected.

5.6 Audit and Incident Response:

Every block event and ledger change is recorded immutably, helping security teams to reconstruct and investigate attacks.

5.7 Proactive Attack Chain Disruption:

By blocking execution before lateral movement, BIDPS limits an adversary's ability to spread from one endpoint to the rest of the network.

Chapter 6

CHALLENGES AND LIMITATIONS

6.1 Fileless and Memory-Resident Attacks

Hash whitelisting cannot see code that exists only in memory. Without memory analysis, only 63.04% of fileless attacks were stopped, and even with Sysmon integration success was 84.7%.

6.2 Whitelist Maintenance Overhead

Every legitimate software change, update or new installation must be registered on the ledger. Initial hashing already takes about 52.83 seconds on the test endpoint and will grow with the number of binaries.

6.3 Ledger Performance and Scalability

The evaluation used a small testbed. Transaction throughput and latency of Hyperledger Fabric under dense, enterprise-scale workloads have not yet been measured.

6.4 Dependence on the Ledger and Gateway

Execution checks require the ledger to be reachable. Network failure or an attack on the ledger infrastructure could delay or block legitimate programs.

6.5 Emulated Evaluation

Results come from emulated attacks on a virtual testbed. Real-world adversaries may adapt to the system, and performance on production networks may differ.

6.6 Privacy and Management Concerns

Continuous monitoring of executions raises privacy questions, and managing certificates, keys and ownership records adds administrative effort.

6.7 Cost and Resource Requirements

Operating a blockchain network, ledger peers and monitoring tools adds infrastructure and expertise costs compared with a conventional antivirus deployment.

Chapter 7

FUTURE SCOPE AND DISCUSSION

7.1 Future Developments

7.1.1 Advanced Memory Integration

Kernel-level event identifiers, such as Sysmon events, can be written directly to the chain so that smart contracts can respond to memory-based attacks automatically.

7.1.2 Large-Scale Evaluation

Testing transaction throughput (TPS) and latency bounds under dense user workloads will show whether the system suits large enterprises.

7.1.3 Stronger Active Mitigation

Mitigation mechanisms can be strengthened to maintain performance under dynamic, complex real-world enterprise conditions.

7.1.4 Behavioural and Machine Learning Analysis

Combining whitelisting with behavioural analytics could improve detection of fileless and zero-day attacks.

7.1.5 Cross-Platform Support

Extending BIDPS to Linux, macOS and mobile endpoints would cover the diverse devices in BYOD environments.

7.1.6 Automated Whitelist Management

Automated approval workflows tied to trusted software vendors and patch pipelines would reduce the maintenance burden.

Chapter 8

CONCLUSION

The present study examines a Blockchain-Enabled Intrusion Detection and Prevention System that closes a key gap in Zero Trust Architecture, the trusted but possibly compromised endpoint. By registering SHA-512 hashes and owners of legitimate binaries on a permissioned Hyperledger Fabric ledger and checking every execution against it, BIDPS turns a passive blockchain log into an active validation and execution gate and shifts trust validation to smart contracts.

Evaluated against 46 MITRE ATT&CK techniques emulated with CALDERA, the system achieved a 100% success rate against file-based attacks and an overall defense success of 95.2% once Sysmon Event ID 8 integration improved fileless detection from 63.04% to 84.7%. The approach halts execution vectors before lateral movement and keeps policy in an immutable source of truth that a compromised node cannot modify. Future research will focus on tighter memory integration, large-scale throughput evaluation and stronger active mitigation.

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APPENDIX

- Reference Paper of the Seminar

  Blockchain-Enabled Intrusion Detection and Prevention System of APTs Within Zero Trust Architecture

- PPT of the Seminar



Department of Computer Science and Engineering
Vidya Academy of Science & Technology
Thalakkottukara, Thrissur - 680 501
(http://www.vidyaacademy.ac.in)